DETAIL CALCULATION BOOKLET

Vessel Name:	M.V. CROWN MARINERS - 67	Operating Route:	Inland & bay crossing
Vessel Type:	Steel Body Cargo Vessel	Plying Time:	Day & Night
Hull Construction:	Single Hull	Statutory Factor kt / kz:	1.40 / 1.40
Length Over All (LOA):	76.21 m	Breadth Moulded (B):	13.41 m
Length BP (LBP):	69.85 m	Depth Moulded (D):	5.18 m
Design Draft (d):	4 m	Block Coeff (Cb):	0.721
Transverse Frame Spacing:	550 mm		

PREPARED & DESIGNED BY	CHECKED & VERIFIED BY
ROUNAK SAHA NILOY Naval Architect and Marine Engineer Designer ShipDyn Ltd.	Amitavo Chowdhury Wye Authorized Person ShipDyn Ltd.

EXECUTIVE STATUTORY COMPLIANCE DASHBOARD

Comprehensive Statutory Scantling Audit organized by 12 Statutory Structural Categories | Construction: Single Hull

GROUP 1: 1. HULL SHELL & DECK PLATING (15 items) — 15 / 15 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
1	Minimum Hull Plate Material	4.00	1.40	5.60	**6.00**	mm	+0.40	PASS
2	Keel Plating Thickness	7.99	1.40	11.19	**12.00**	mm	+0.81	PASS
3	Keel Plating Width	1099.20	1.40	1099.20	**1100.00**	mm	+0.80	PASS
4	Bottom Shell Plating	7.49	1.40	10.49	**11.00**	mm	+0.51	PASS
5	Bilge Plating Thickness	9.49	1.40	13.29	**14.00**	mm	+0.71	PASS
6	Side Shell Plating	7.49	1.40	10.49	**12.00**	mm	+1.51	PASS
7	Sheer Strake Plating	7.49	1.40	10.49	**11.00**	mm	+0.51	PASS
8	Main Strength Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
9	Poop Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
10	Forecastle Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
11	Second / Tween Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
12	Navigation Bridge Deck Plating	5.99	1.40	8.39	**9.00**	mm	+0.61	PASS
13	Strength Deck at Hatchways	8.99	1.40	12.59	**13.00**	mm	+0.41	PASS
14	Plate Stem Plating	10.98	1.40	15.38	**16.00**	mm	+0.62	PASS
15	Transom Stern Plating	7.49	1.40	10.49	**11.00**	mm	+0.51	PASS

GROUP 2: 2. STEMS, STERN FRAMES & RUDDER (16 items) — 16 / 16 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
16	Bar Stem Breadth & Thickness	115.00	1.00	115.00	**115.00**	mm	+0.00	PASS
17	Stern Frame Bar Breadth	115.00	1.00	115.00	**115.00**	mm	+0.00	PASS
18	Propeller Post Dimensions	130.00	1.00	130.00	**130.00**	mm	+0.00	PASS
19	Sole Piece Breadth	150.00	1.00	150.00	**150.00**	mm	+0.00	PASS
20	Sole Piece Thickness	100.00	1.00	100.00	**100.00**	mm	+0.00	PASS
21	Propeller Boss Wall Thickness	40.00	1.00	40.00	**40.00**	mm	+0.00	PASS
22	V-Strut Propeller Shaft Bracket	45.00	1.00	45.00	**45.00**	mm	+0.00	PASS
23	I-Strut Propeller Shaft Bracket	50.00	1.00	50.00	**50.00**	mm	+0.00	PASS
24	Rudder Stock Diameter - Supported	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
25	Rudder Stock Diameter - Spade	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
26	Rudder Pintle Diameter	193.80	1.00	193.80	**193.80**	mm	+0.00	PASS
27	Rudder Coupling Flange Thickness	28.00	1.00	28.00	**28.00**	mm	+0.00	PASS
28	Rudder Coupling Bolts Diameter	25.00	1.00	25.00	**25.00**	mm	+0.00	PASS
29	Rudder Blade Plating Thickness	8.00	1.40	11.20	**12.00**	mm	+0.80	PASS
30	Steering Tiller / Quadrant Hub	110.00	1.00	110.00	**110.00**	mm	+0.00	PASS
31	Steering Chains & Shackles	16.00	1.00	16.00	**16.00**	mm	+0.00	PASS

GROUP 3: 3. BOTTOM CONSTRUCTION, FLOORS & KEELSONS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
32	Centre Keelson	9.49	1.40	13.29	**1274.33**	cm3	+1261.04	PASS
33	Side Keelson	9.49	1.40	13.29	**686.93**	cm3	+673.64	PASS
34	Bottom Transverse Floor Plate	1645.99	1.40	2304.38	**2394.23**	cm3	+89.85	PASS

GROUP 4: 4. FRAMING SYSTEMS & STRINGERS (5 items) — 5 / 5 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
35	Main Side Transverse Frame	240.20	1.40	336.28	**341.76**	cm3	+5.48	PASS
36	Side Web Frame	238.14	1.40	333.40	**864.00**	cm3	+530.60	PASS
37	Side Stringer	238.14	1.40	333.40	**432.49**	cm3	+99.09	PASS
38	Bilge Bracket	450.00	1.00	450.00	**450.00**	mm	+0.00	PASS
39	Deck Beam End Bracket to Frame	350.00	1.00	350.00	**350.00**	mm	+0.00	PASS

GROUP 5: 5. DECKS, GIRDERS & PILLARS (11 items) — 11 / 11 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
40	Solid Round Deck Pillar	57.00	1.00	57.00	**57.00**	mm	+0.00	PASS
41	Tubular Steel Deck Pillar	114.30	1.00	114.30	**114.30**	mm	+0.00	PASS
42	Transverse Strength Deck Beam	10.77	1.40	15.08	**128.14**	cm3	+113.06	PASS
43	Strength Deck Longitudinal Beams	10.77	1.40	15.08	**128.14**	cm3	+113.06	PASS
44	Strength Deck Transverse Web Beams	549.58	1.40	769.41	**837.73**	cm3	+68.32	PASS
45	Main Deck Center Longitudinal Girder	549.58	1.40	769.41	**1440.84**	cm3	+671.43	PASS
46	Superstructure Roof Beam	55.00	1.40	77.00	**86.83**	cm3	+9.83	PASS
47	Side Roof Girder	120.00	1.40	168.00	**182.20**	cm3	+14.20	PASS
48	Centre Roof Girder	150.00	1.40	210.00	**233.43**	cm3	+23.43	PASS
49	Superstructure Wall Stiffener	16.00	1.40	22.40	**71.84**	cm3	+49.44	PASS
50	Superstructure Main Frame	16.00	1.40	22.40	**118.29**	cm3	+95.89	PASS



GROUP 6: 6. WATERTIGHT BULKHEADS & TANKS (5 items) — 5 / 5 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
51	Collision Bulkhead Position	3.49	1.00	3.49	**4.80**	m	+1.31	PASS
52	Transverse Watertight Bulkhead Plating	6.68	1.40	9.35	**10.00**	mm	+0.65	PASS
53	Collision & Aft Peak Bulkhead Plating	7.68	1.40	10.75	**11.00**	mm	+0.25	PASS
54	Bow Chain Locker Wall Plating	6.98	1.40	6.98	**8.00**	mm	+1.02	PASS
55	Watertight Bulkhead Vertical Stiffeners	248.16	1.40	347.42	**370.93**	cm3	+23.51	PASS

GROUP 7: 7. HATCHWAYS, COAMINGS & COVERS (7 items) — 7 / 7 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
56	Longitudinal Hatch Coaming Plating	8.99	1.40	12.59	**14.00**	mm	+1.41	PASS
57	Transverse Hatch Coaming Plating	8.99	1.40	12.59	**14.00**	mm	+1.41	PASS
58	Hatch Coaming Stay Bracket	300.00	1.00	300.00	**300.00**	mm	+0.00	PASS
59	Cargo Hatchway Transverse Beams	13.00	1.40	18.20	**871.68**	cm3	+853.48	PASS
60	Cargo Hatchway Side Longitudinal Beams	13.00	1.40	18.20	**1273.59**	cm3	+1255.39	PASS
61	Portable Removable Hatch Beam	13.00	1.40	18.20	**592.79**	cm3	+574.59	PASS
62	Steel Hatch Cover Stiffeners	5.00	1.40	7.00	**78.77**	cm3	+71.77	PASS

GROUP 8: 8. MACHINERY CASINGS & SEATINGS (3 items) — 0 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
63	Machinery Casing Plating	5.00	1.40	7.00	**5.00**	mm	-2.00	FAIL
64	Engine Foundation Top Plate	19.00	1.40	26.60	**19.00**	mm	-7.60	FAIL
65	Engine Foundation Longitudinal Girders	12.00	1.40	16.80	**12.00**	mm	-4.80	FAIL

GROUP 9: 9. BULWARKS & FREEING PORTS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
66	Bulwark Minimum Height	1000.00	1.00	1000.00	**1000.00**	mm	+0.00	PASS
67	Bulwark Plating Thickness	5.99	1.40	5.99	**6.00**	mm	+0.01	PASS
68	Freeing Port Total Area per Side	4.54	1.00	4.54	**4.54**	m2	+0.00	PASS

GROUP 10: 10. EQUIPMENT NUMBER, ANCHORS & CABLES (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
69	Equipment Number Z Calculation	323.30	1.00	323.30	**323.30**	Z	+0.00	PASS
70	Bower Anchor Mass	570.00	1.00	570.00	**570.00**	kg	+0.00	PASS
71	Stud Link Chain Cable Total Length	275.00	1.00	275.00	**275.00**	m	+0.00	PASS

GROUP 11: 11. MACHINERY SHAFTING & BEARINGS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
72	Intermediate Shaft Diameter	85.00	1.00	85.00	**85.00**	mm	+0.00	PASS
73	Thrust Shaft Diameter	95.00	1.00	95.00	**95.00**	mm	+0.00	PASS
74	Propeller Shaft Diameter	105.00	1.00	105.00	**105.00**	mm	+0.00	PASS

GROUP 12: 12. PIPING & PUMPING SYSTEMS (3 items) — 3 / 3 COMPLIED

No.	Scantling Component Description	Baseline	Factor	Required	Provided	Unit	Margin	Status
75	Main Bilge Suction Pipe Diameter	65.00	1.00	65.00	**65.00**	mm	+0.00	PASS
76	Branch Bilge Suction Pipe Diameter	50.00	1.00	50.00	**50.00**	mm	+0.00	PASS
77	Bilge & Ballast Pipe Wall Thickness	4.50	1.00	4.50	**4.50**	mm	+0.00	PASS



SHEET 01 : CENTRE KEELSON | T 500x100x10 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	11.0	6050.0
Stiffener Web	500.0	12.0	6000.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**13050.0**

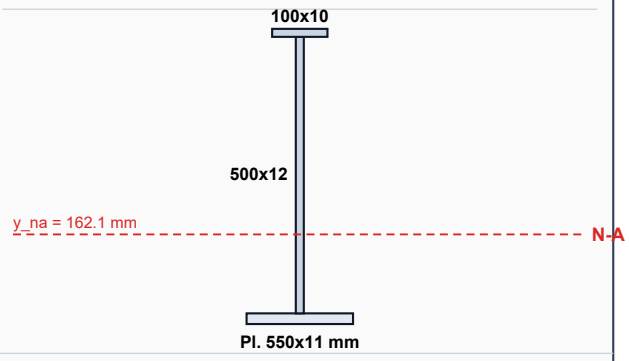
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	162.090 mm
Plate Parallel-Axis Inertia (Ip):	148409666.77 mm⁴
Web Parallel-Axis Inertia (Iw):	183699083.12 mm⁴
Flange Parallel-Axis Inertia (If):	125260594.31 mm⁴
Combined Section Inertia (I_NA):	**457369344.20 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**1274.33 cm³ (1274329 mm³)**
Rule Statutory Required (Table 16):	13.29 cm³ (kz = 1.40)

CENTRE KEELSON

Profile: T 500x100x12 mm

Section Modulus: Z = 1274.33 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 02 : SIDE KEELSON | T 350x100x10 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	11.0	6050.0
Stiffener Web	350.0	10.0	3500.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**10550.0**

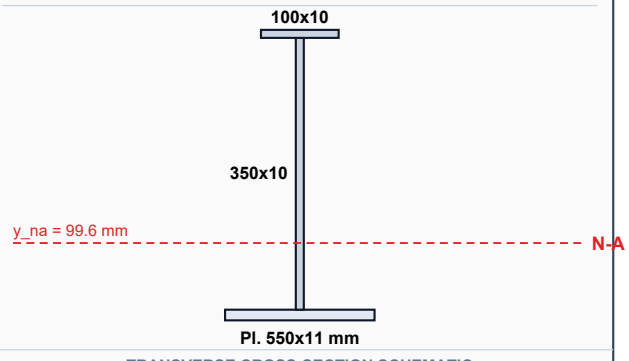
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	99.552 mm
Plate Parallel-Axis Inertia (Ip):	53578116.34 mm⁴
Web Parallel-Axis Inertia (Iw):	61885484.83 mm⁴
Flange Parallel-Axis Inertia (If):	71002799.32 mm⁴
Combined Section Inertia (I_NA):	**186466400.49 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**686.93 cm³ (686933 mm³)**
Rule Statutory Required (Table 16):	13.29 cm³ (kz = 1.40)

SIDE KEELSON

Profile: T 350x100x10 mm

Section Modulus: Z = 686.93 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 03 : BOTTOM TRANSVERSE FLOOR PLATE | T 500x180x20 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	11.0	6050.0
Stiffener Web	500.0	12.0	6000.0
Stiffener Flange	180.0	20.0	3600.0
Total Section Area			**15650.0**

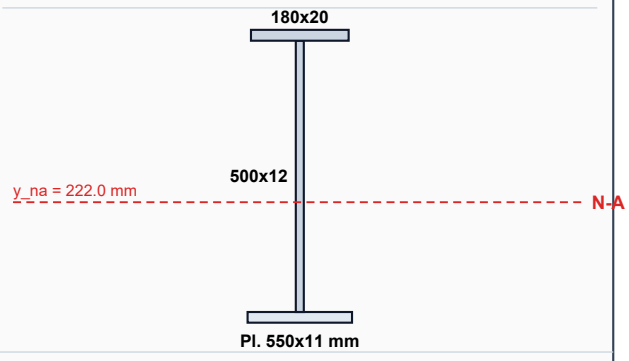
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	222.037 mm
Plate Parallel-Axis Inertia (Ip):	283734373.96 mm⁴
Web Parallel-Axis Inertia (Iw):	134108813.21 mm⁴
Flange Parallel-Axis Inertia (If):	321884508.37 mm⁴
Combined Section Inertia (I_NA):	**739727695.54 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**2394.23 cm³ (2394225 mm³)**
Rule Statutory Required (Table 13):	2304.38 cm³ (kz = 1.40)

BOTTOM TRANSVERSE FLOOR PLATE

Profile: T 500x180x12 mm

Section Modulus: Z = 2394.23 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 04 : MAIN SIDE TRANSVERSE FRAME | L 200x100x10 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	200.0	12.0	2400.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**10000.0**

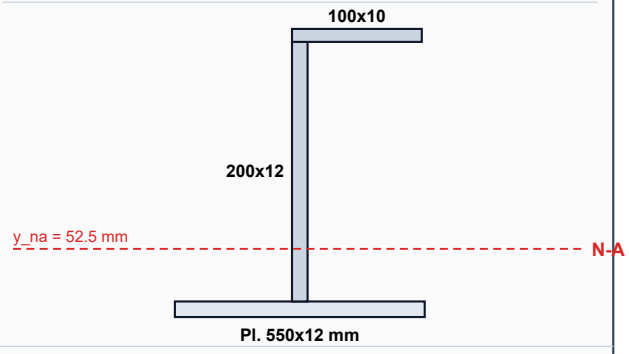
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	52.540 mm
Plate Parallel-Axis Inertia (Ip):	14374612.56 mm⁴
Web Parallel-Axis Inertia (Iw):	16485179.84 mm⁴
Flange Parallel-Axis Inertia (If):	27055424.93 mm⁴
Combined Section Inertia (I_NA):	**57915217.33 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**341.76 cm³ (341763 mm³)**
Rule Statutory Required (Table 17):	336.28 cm³ (kz = 1.40)

MAIN SIDE TRANSVERSE FRAME

Profile: L 200x100x12 mm

Section Modulus: Z = 341.76 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 05 : SIDE WEB FRAME | T 350x120x12 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	350.0	11.0	3850.0
Stiffener Flange	120.0	12.0	1440.0
Total Section Area			**11890.0**

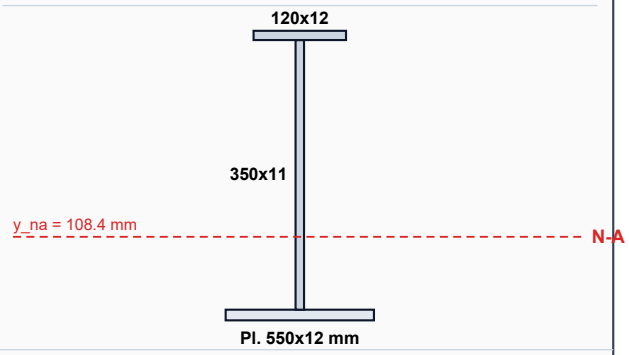
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	108.450 mm
Plate Parallel-Axis Inertia (Ip):	69352759.63 mm⁴
Web Parallel-Axis Inertia (Iw):	63057003.39 mm⁴
Flange Parallel-Axis Inertia (If):	97024643.03 mm⁴
Combined Section Inertia (I_NA):	**229434406.06 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**864.00 cm³ (863997 mm³)**
Rule Statutory Required (Table 19):	333.40 cm³ (kz = 1.40)

SIDE WEB FRAME

Profile: T 350x120x11 mm

Section Modulus: Z = 864.00 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 06 : SIDE STRINGER | T 250x100x10 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	250.0	10.0	2500.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**10100.0**

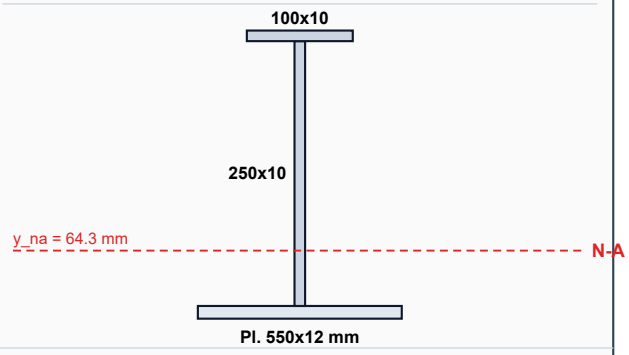
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	64.267 mm
Plate Parallel-Axis Inertia (Ip):	22486737.01 mm⁴
Web Parallel-Axis Inertia (Iw):	26245937.73 mm⁴
Flange Parallel-Axis Inertia (If):	41108870.14 mm⁴
Combined Section Inertia (I_NA):	**89841544.88 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**432.49 cm³ (432486 mm³)**
Rule Statutory Required (Table 19):	333.40 cm³ (kz = 1.40)

SIDE STRINGER

Profile: T 250x100x10 mm

Section Modulus: Z = 432.49 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 07 : TRANSVERSE STRENGTH DECK BEAM | L 125x75x9 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	125.0	9.0	1125.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**6750.0**

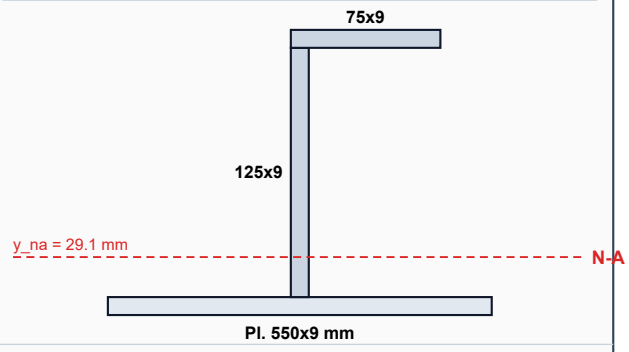
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	29.067 mm
Plate Parallel-Axis Inertia (Ip):	3020842.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3490505.00 mm⁴
Flange Parallel-Axis Inertia (If):	8088123.00 mm⁴
Combined Section Inertia (I_NA):	**14599470.00 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**128.14 cm³ (128140 mm³)**
Rule Statutory Required (Table 20):	15.08 cm³ (kz = 1.40)

TRANSVERSE STRENGTH DECK BEAM

Profile: L 125x75x9 mm

Section Modulus: Z = 128.14 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 08 : STRENGTH DECK LONGITUDINAL BEAMS | L 125x75x9 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	125.0	9.0	1125.0
Stiffener Flange	75.0	9.0	675.0
Total Section Area			**6750.0**

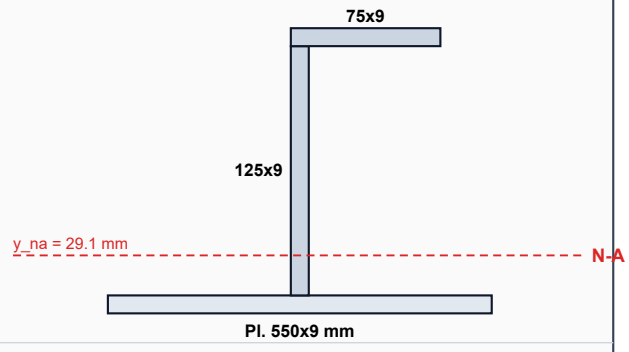
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	29.067 mm
Plate Parallel-Axis Inertia (Ip):	3020842.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3490505.00 mm⁴
Flange Parallel-Axis Inertia (If):	8088123.00 mm⁴
Combined Section Inertia (I_NA):	**14599470.00 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**128.14 cm³ (128140 mm³)**
Rule Statutory Required (Table 20):	15.08 cm³ (kz = 1.40)

STRENGTH DECK LONGITUDINAL BEAMS

Profile: L 125x75x9 mm

Section Modulus: Z = 128.14 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 09 : STRENGTH DECK TRANSVERSE WEB BEAMS | T 300x150x14 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	300.0	10.0	3000.0
Stiffener Flange	150.0	14.0	2100.0
Total Section Area			**10050.0**

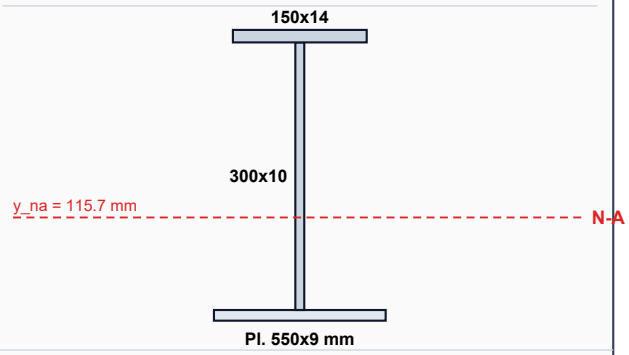
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	115.709 mm
Plate Parallel-Axis Inertia (Ip):	61252199.52 mm⁴
Web Parallel-Axis Inertia (Iw):	28122343.67 mm⁴
Flange Parallel-Axis Inertia (If):	84278955.50 mm⁴
Combined Section Inertia (I_NA):	**173653498.69 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**837.73 cm³ (837728 mm³)**
Rule Statutory Required (Table 21):	769.41 cm³ (kz = 1.40)

STRENGTH DECK TRANSVERSE WEB B..

Profile: T 300x150x10 mm

Section Modulus: Z = 837.73 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 10 : MAIN DECK CENTER LONGITUDINAL GIRDER | T 450x150x14 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	450.0	11.0	4950.0
Stiffener Flange	150.0	14.0	2100.0
Total Section Area			**12000.0**

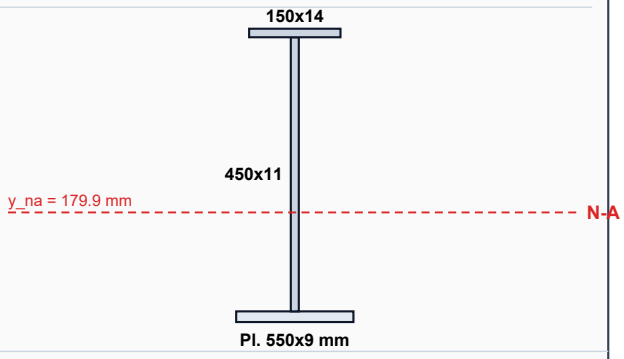
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	179.931 mm
Plate Parallel-Axis Inertia (Ip):	152375223.71 mm⁴
Web Parallel-Axis Inertia (Iw):	98002227.15 mm⁴
Flange Parallel-Axis Inertia (If):	171888492.43 mm⁴
Combined Section Inertia (I_NA):	**422265943.28 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**1440.84 cm³ (1440843 mm³)**
Rule Statutory Required (Table 21):	769.41 cm³ (kz = 1.40)

MAIN DECK CENTER LONGITUDINAL ..

Profile: T 450x150x11 mm

Section Modulus: Z = 1440.84 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 11 : SUPERSTRUCTURE ROOF BEAM | L 100x75x8 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	100.0	8.0	800.0
Stiffener Flange	75.0	8.0	600.0
Total Section Area			**6350.0**

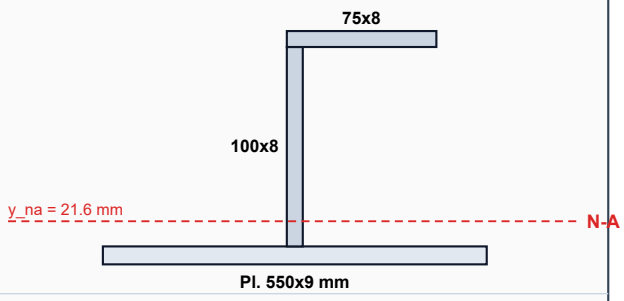
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	21.618 mm
Plate Parallel-Axis Inertia (Ip):	1483909.51 mm⁴
Web Parallel-Axis Inertia (Iw):	1784591.21 mm⁴
Flange Parallel-Axis Inertia (If):	5013589.87 mm⁴
Combined Section Inertia (I_NA):	**8282090.58 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**86.83 cm³ (86831 mm³)**
Rule Statutory Required (Table 36):	77.00 cm³ (kz = 1.40)

SUPERSTRUCTURE ROOF BEAM

Profile: L 100x75x8 mm

Section Modulus: Z = 86.83 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 12 : SIDE ROOF GIRDER | L 150x90x9 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	150.0	9.0	1350.0
Stiffener Flange	90.0	9.0	810.0
Total Section Area			**7110.0**

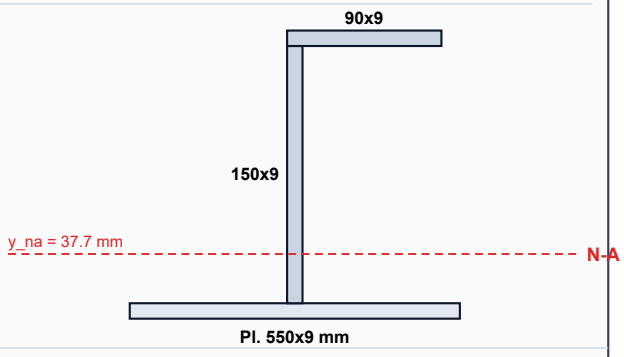
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	37.709 mm
Plate Parallel-Axis Inertia (Ip):	5492413.24 mm⁴
Web Parallel-Axis Inertia (Iw):	5424123.92 mm⁴
Flange Parallel-Axis Inertia (If):	12822430.18 mm⁴
Combined Section Inertia (I_NA):	**23738967.34 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**182.20 cm³ (182199 mm³)**
Rule Statutory Required (Table 37):	168.00 cm³ (kz = 1.40)

SIDE ROOF GIRDER

Profile: L 150x90x9 mm

Section Modulus: Z = 182.20 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 13 : CENTRE ROOF GIRDER | L 160x100x10 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	160.0	10.0	1600.0
Stiffener Flange	100.0	10.0	1000.0
Total Section Area			**7550.0**

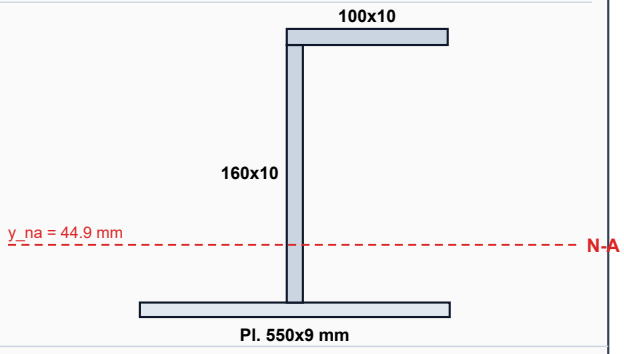
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	44.858 mm
Plate Parallel-Axis Inertia (Ip):	8095661.45 mm⁴
Web Parallel-Axis Inertia (Iw):	6531013.45 mm⁴
Flange Parallel-Axis Inertia (If):	16686088.71 mm⁴
Combined Section Inertia (I_NA):	**31312763.60 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**233.43 cm³ (233429 mm³)**
Rule Statutory Required (Table 37):	210.00 cm³ (kz = 1.40)

CENTRE ROOF GIRDER

Profile: L 160x100x10 mm

Section Modulus: Z = 233.43 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 14 : SUPERSTRUCTURE WALL STIFFENER | L 100x65x7 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	100.0	7.0	700.0
Stiffener Flange	65.0	7.0	455.0
Total Section Area			**7755.0**

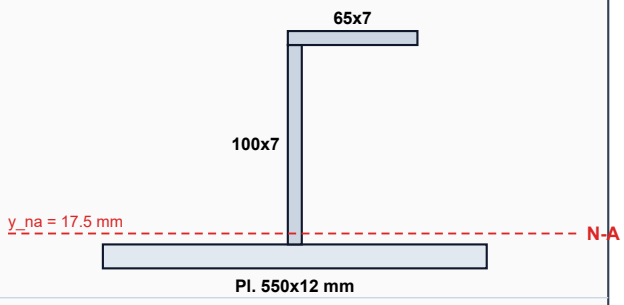
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	17.479 mm
Plate Parallel-Axis Inertia (Ip):	948920.89 mm⁴
Web Parallel-Axis Inertia (Iw):	1970794.00 mm⁴
Flange Parallel-Axis Inertia (If):	4373518.06 mm⁴
Combined Section Inertia (I_NA):	**7293232.95 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**71.84 cm³ (71840 mm³)**
Rule Statutory Required (Table 35):	22.40 cm³ (kz = 1.40)

SUPERSTRUCTURE WALL STIFFENER

Profile: L 100x65x7 mm

Section Modulus: Z = 71.84 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 15 : SUPERSTRUCTURE MAIN FRAME | L 125x75x8 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	12.0	6600.0
Stiffener Web	125.0	8.0	1000.0
Stiffener Flange	75.0	8.0	600.0
Total Section Area			**8200.0**

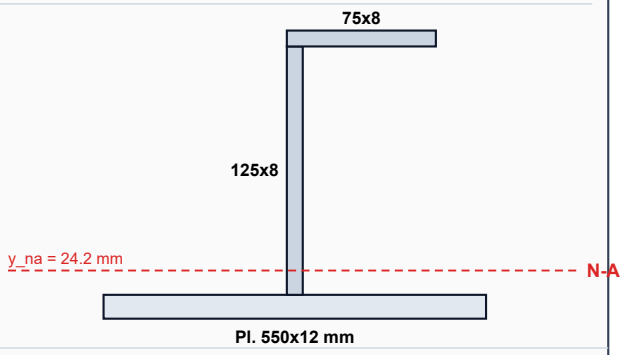
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	24.232 mm
Plate Parallel-Axis Inertia (Ip):	2273008.00 mm⁴
Web Parallel-Axis Inertia (Iw):	3828984.58 mm⁴
Flange Parallel-Axis Inertia (If):	8184100.51 mm⁴
Combined Section Inertia (I_NA):	**14286093.09 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**118.29 cm³ (118293 mm³)**
Rule Statutory Required (Table 35):	22.40 cm³ (kz = 1.40)

SUPERSTRUCTURE MAIN FRAME

Profile: L 125x75x8 mm

Section Modulus: Z = 118.29 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 16 : WATERTIGHT BULKHEAD VERTICAL STIFFENERS | L 200x100x12 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	10.0	5500.0
Stiffener Web	200.0	12.0	2400.0
Stiffener Flange	100.0	12.0	1200.0
Total Section Area			**9100.0**

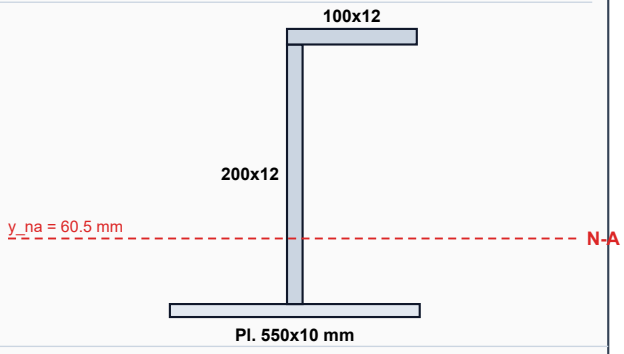
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	60.516 mm
Plate Parallel-Axis Inertia (Ip):	16997273.01 mm⁴
Web Parallel-Axis Inertia (Iw):	13876684.17 mm⁴
Flange Parallel-Axis Inertia (If):	29024548.68 mm⁴
Combined Section Inertia (I_NA):	**59898505.86 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**370.93 cm³ (370926 mm³)**
Rule Statutory Required (Table 26):	347.42 cm³ (kz = 1.40)

WATERTIGHT BULKHEAD VERTICAL S..

Profile: L 200x100x12 mm

Section Modulus: Z = 370.93 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 17 : CARGO HATCHWAY TRANSVERSE BEAMS | T 350x120x12 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	13.0	7150.0
Stiffener Web	350.0	11.0	3850.0
Stiffener Flange	120.0	12.0	1440.0
Total Section Area			**12440.0**

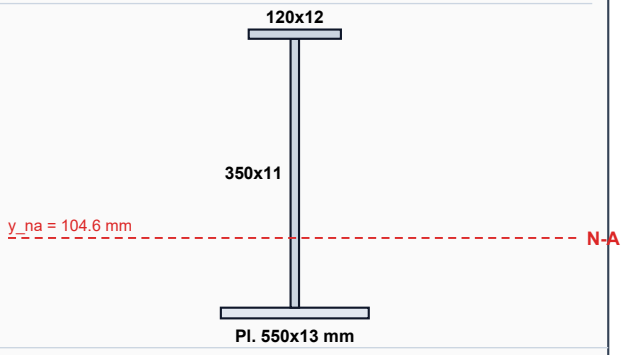
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	104.633 mm
Plate Parallel-Axis Inertia (Ip):	68955862.66 mm⁴
Web Parallel-Axis Inertia (Iw):	66059776.82 mm⁴
Flange Parallel-Axis Inertia (If):	100658722.01 mm⁴
Combined Section Inertia (I_NA):	**235674361.49 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**871.68 cm³ (871683 mm³)**
Rule Statutory Required (Table 32):	18.20 cm³ (kz = 1.40)

CARGO HATCHWAY TRANSVERSE BEAMS

Profile: T 350x120x11 mm

Section Modulus: Z = 871.68 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 18 : CARGO HATCHWAY SIDE LONGITUDINAL BEAMS | T 400x140x14 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	13.0	7150.0
Stiffener Web	400.0	12.0	4800.0
Stiffener Flange	140.0	14.0	1960.0
Total Section Area			**13910.0**

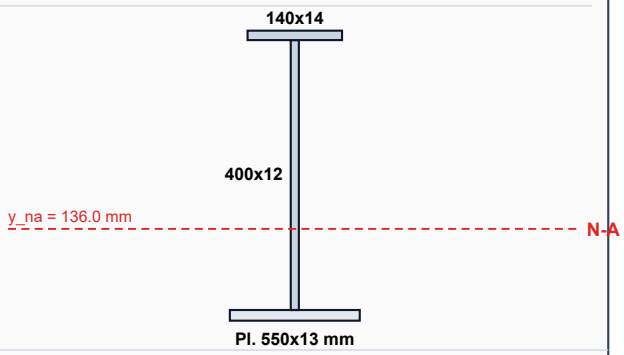
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	136.023 mm
Plate Parallel-Axis Inertia (Ip):	120049923.21 mm⁴
Web Parallel-Axis Inertia (Iw):	92442462.85 mm⁴
Flange Parallel-Axis Inertia (If):	158092563.47 mm⁴
Combined Section Inertia (I_NA):	**370584949.53 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**1273.59 cm³ (1273587 mm³)**
Rule Statutory Required (Table 32):	18.20 cm³ (kz = 1.40)

CARGO HATCHWAY SIDE LONGITUDIN..

Profile: T 400x140x12 mm

Section Modulus: Z = 1273.59 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.

SHEET 19 : PORTABLE REMOVABLE HATCH BEAM | T 300x100x12 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	300.0	10.0	3000.0
Stiffener Flange	100.0	12.0	1200.0
Total Section Area			**9150.0**

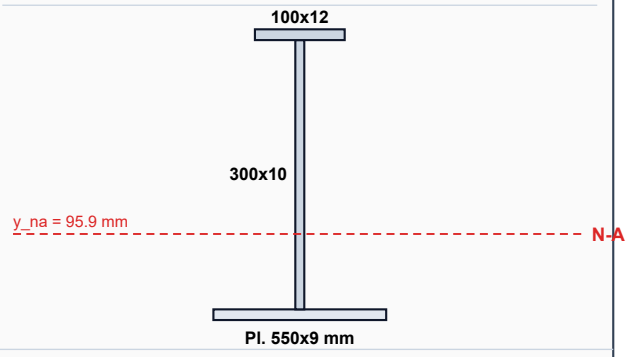
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	95.877 mm
Plate Parallel-Axis Inertia (Ip):	41364749.83 mm⁴
Web Parallel-Axis Inertia (Iw):	34453520.76 mm⁴
Flange Parallel-Axis Inertia (If):	57632241.09 mm⁴
Combined Section Inertia (I_NA):	**133450511.68 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**592.79 cm³ (592789 mm³)**
Rule Statutory Required (Table 32):	18.20 cm³ (kz = 1.40)

PORTABLE REMOVABLE HATCH BEAM

Profile: T 300x100x10 mm

Section Modulus: Z = 592.79 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

SHEET 20 : STEEL HATCH COVER STIFFENERS | L 100x65x8 mm

STATUS: COMPLIANT

Item / Element	Width b [mm]	Thk t [mm]	Area a [mm²]
Attached Plate	550.0	9.0	4950.0
Stiffener Web	100.0	8.0	800.0
Stiffener Flange	65.0	8.0	520.0
Total Section Area			**6270.0**

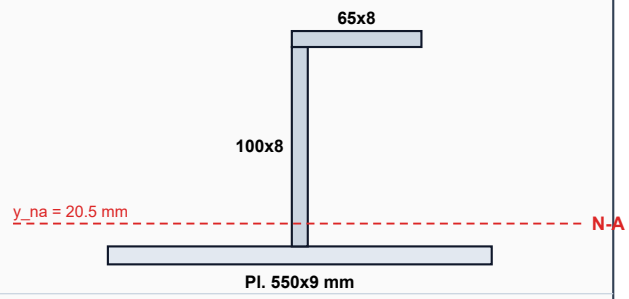
Centroid & Moments of Inertia	Value & Unit
Neutral Axis Height from Plate Base (y_na):	20.452 mm
Plate Parallel-Axis Inertia (Ip):	1293044.88 mm⁴
Web Parallel-Axis Inertia (Iw):	1855415.87 mm⁴
Flange Parallel-Axis Inertia (If):	4456627.39 mm⁴
Combined Section Inertia (I_NA):	**7605088.15 mm⁴**

SECTION MODULUS & COMPLIANCE EVALUATION	
Actual Section Modulus:	**78.77 cm³ (78770 mm³)**
Rule Statutory Required (Table 34):	7.00 cm³ (kz = 1.40)

STEEL HATCH COVER STIFFENERS

Profile: L 100x65x8 mm

Section Modulus: Z = 78.77 cm³



TRANSVERSE CROSS-SECTION SCHEMATIC

ROUNAK SAHA NILOY
Naval Architect and Marine Engineer
Designer
ShipDyn Ltd.